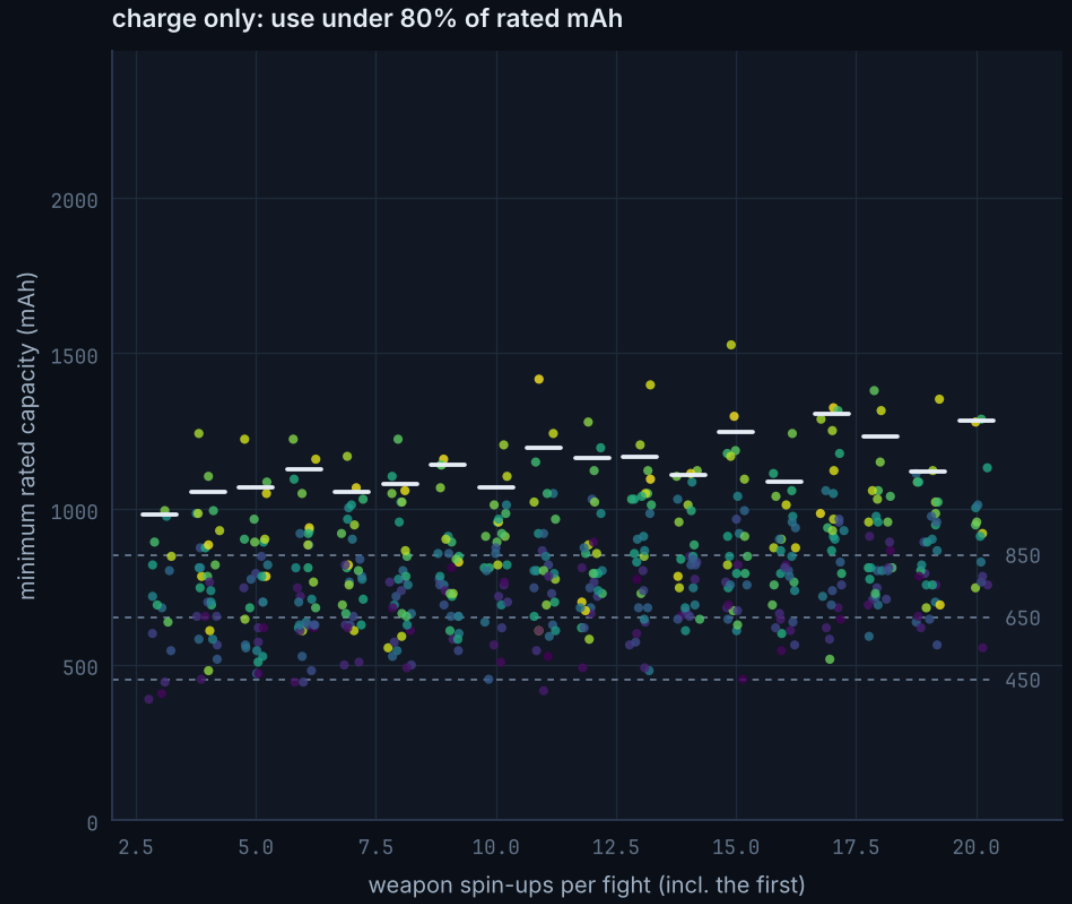
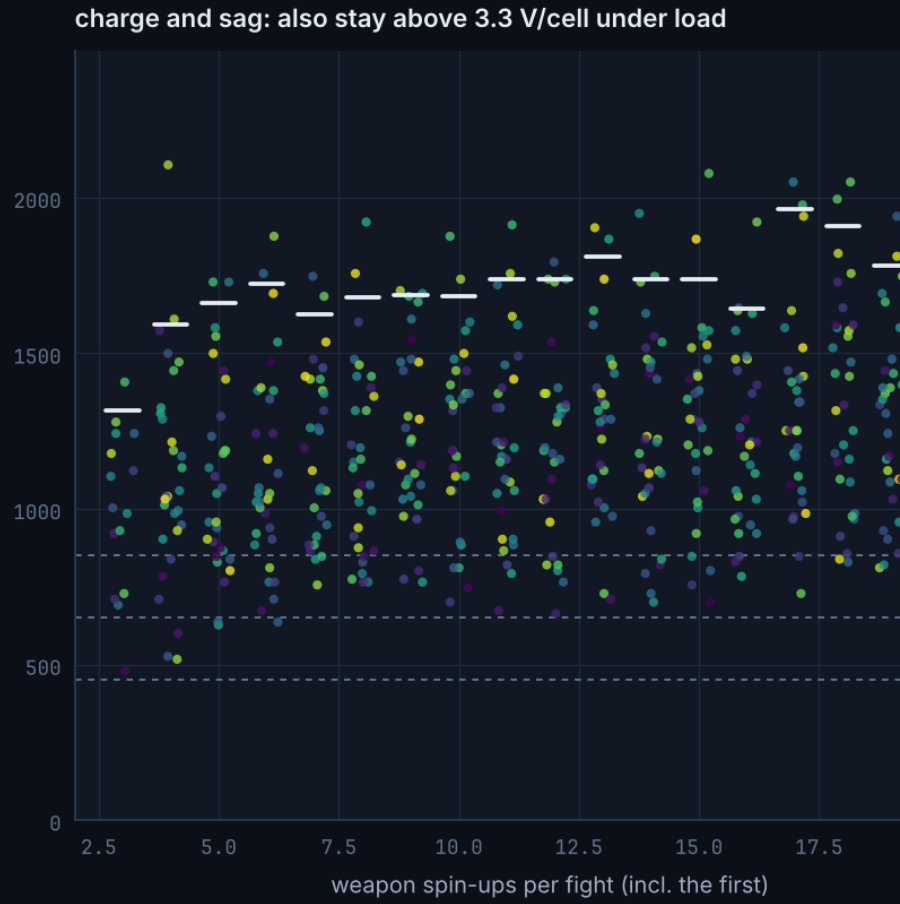
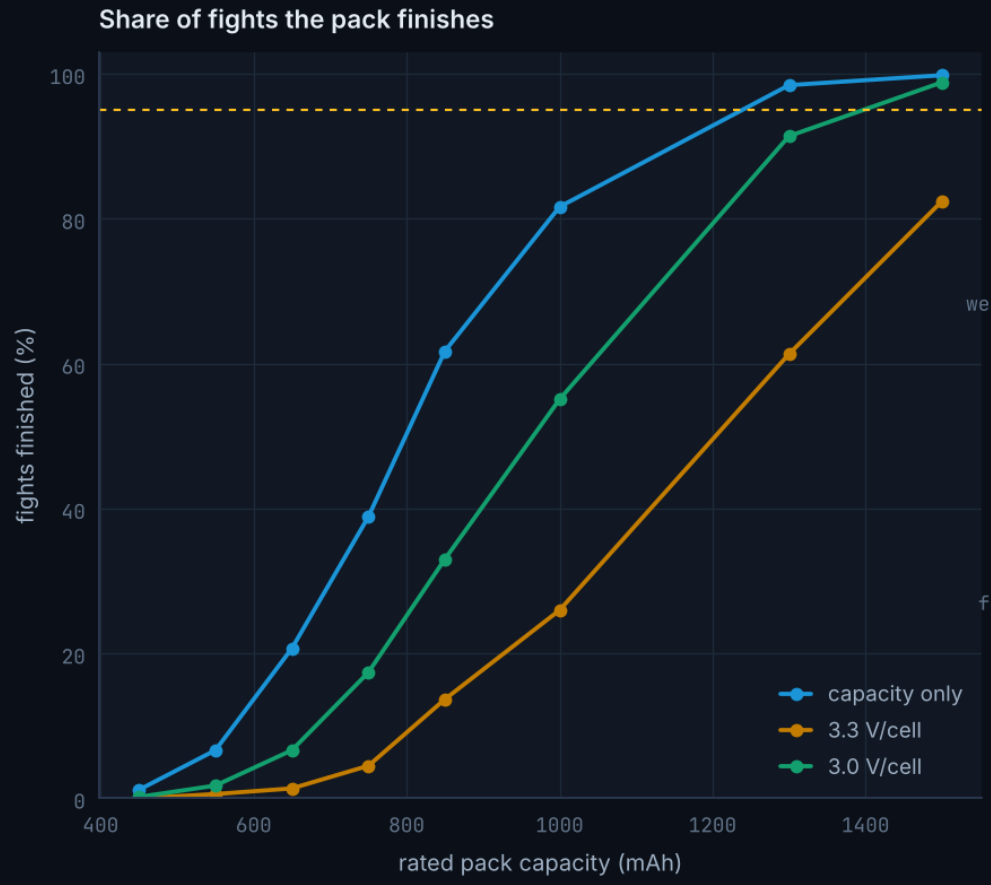
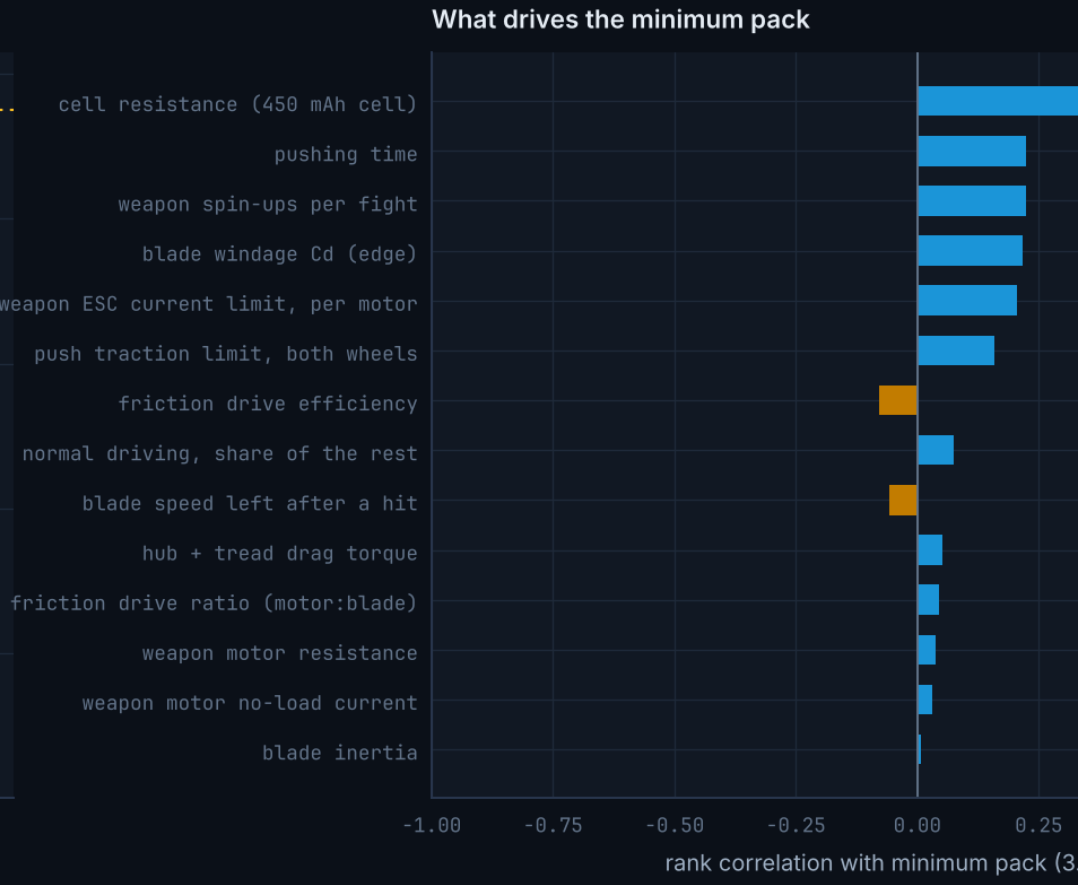


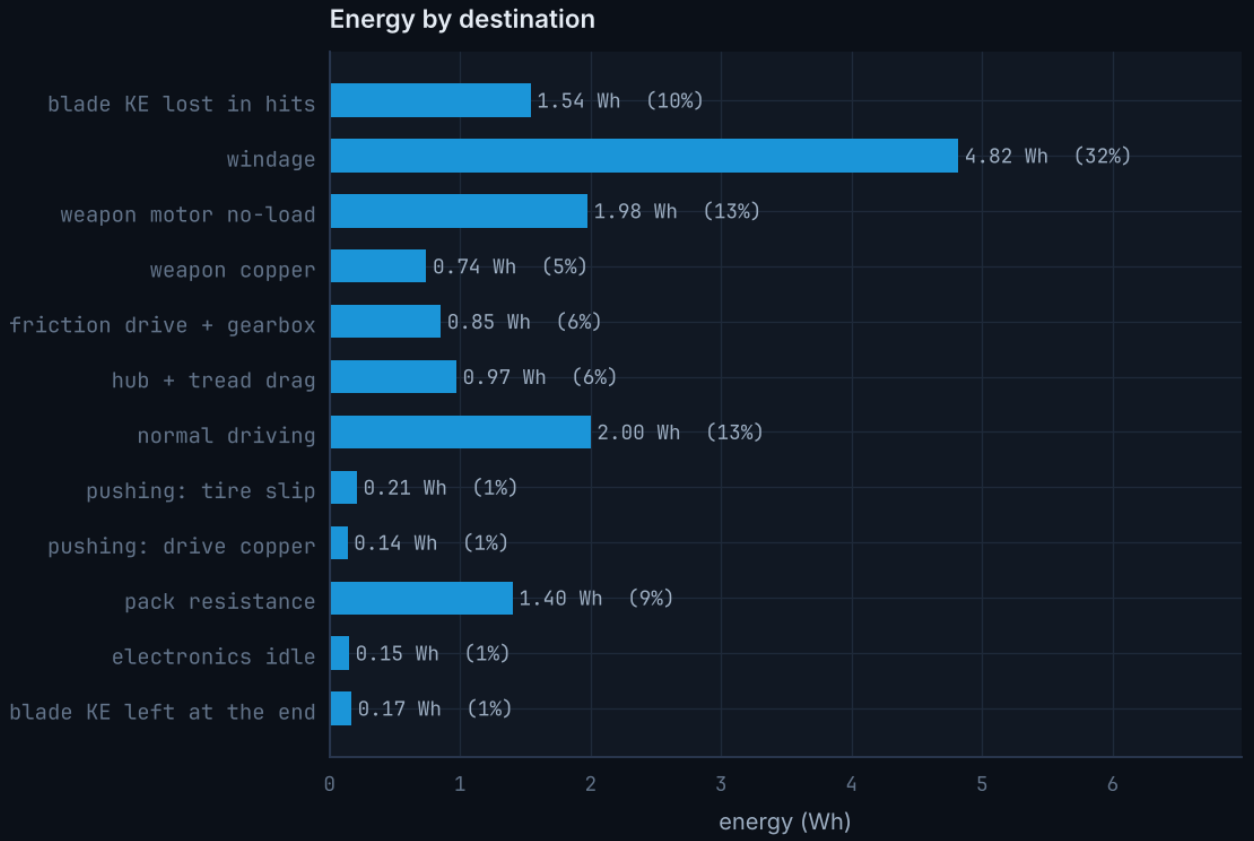


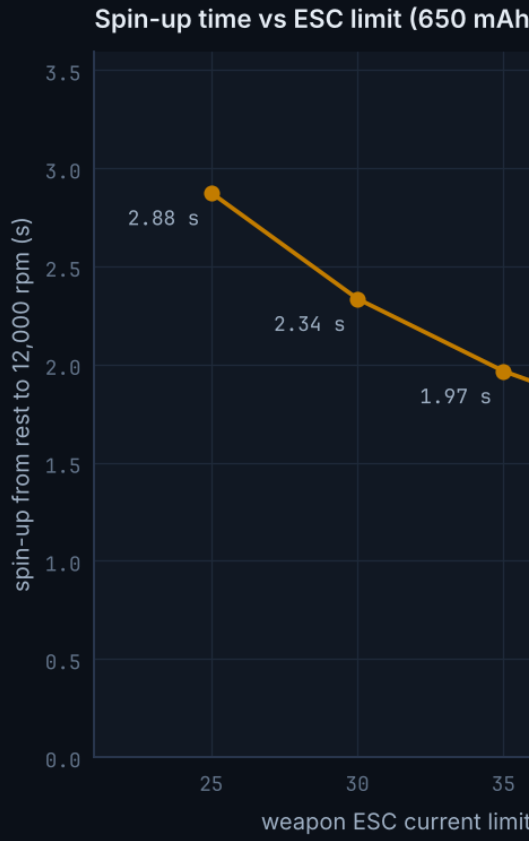
White bars: 95th percentile of the minimum pack for that spin-up count. Dashed lines: common 6S packs (mAh).



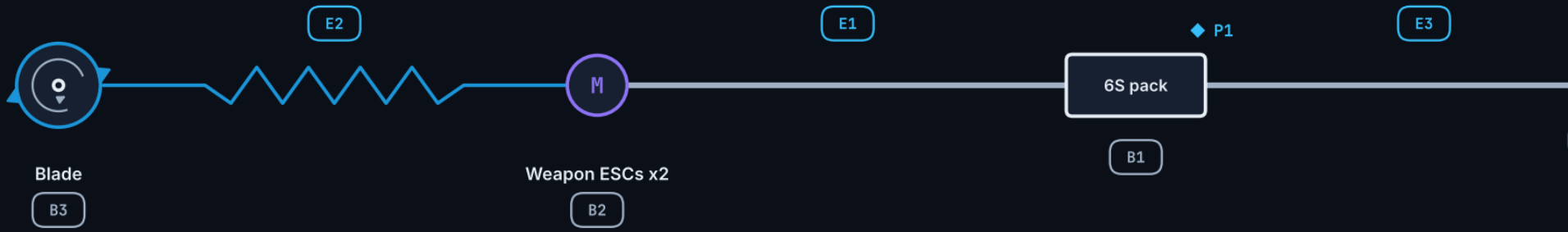


Blue: more of this needs a bigger pack. Orange: more of this needs a smaller pack. Dashed: 95% of fights.





Spin-up is current-limited the whole way: back-EMF at 12,000 rpm blade speed is about half the pack voltage.



KEY

spring · spring + slider · contact · damper · rigid link · plastic hinge · probe · E1 see table

PARAMETERS

B1 6S pack
OCV(SOC) + R0 + R1||C1
R0 scales with 1/capacity

B2 Weapon ESCs x2
2807 1350 KV
current limit 25-45 A

B3 Blade
J 7.6-8.8e-4 kg m^2
12,000 rpm target

B4 Drive ESCs x2
Compact V2 (1806 2300 KV)
20 A limit

E1 Bus
constant-power load
src: ecm_constant_power

E2 Friction drive
G 1.25-1.5, eta 0.75-0.95
src: esc_supply

E3 Bus
drive power trace / push
src: motor_current

E4 Tires
push traction 8-150 N
src: coulomb_tanh

P1 charge used, loaded voltage
measured on 6S pack
src: failure check